

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 18

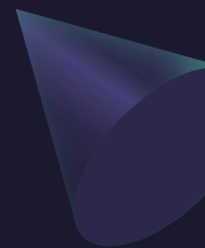


Any Reminders? Any Questions?

- Exam on Friday (discussion) covers 15.1-15.3 + 15.6-15.9
- See email message with resources
 - SP26 Practice Exam/Problems!
- Review during Thursday's class
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



ALVARO: Start the recording!



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Review of Double and Triple Integrals



Example:

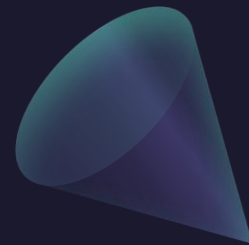
1. **Change of Order of Integration:** Consider the double integral of a continuous function $f(x, y)$ over the region D bounded by $y = x^2$, $y = 9$, and the y -axis in the first quadrant. Set up the integral in both orders: $dy dx$ and $dx dy$.

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Example:

2. **Polar Coordinates:** Evaluate $\iint_D \frac{1}{\sqrt{x^2+y^2+1}} dA$ where D is the region in the first quadrant bounded by the circle $x^2 + y^2 = 16$ and the x and y axes.

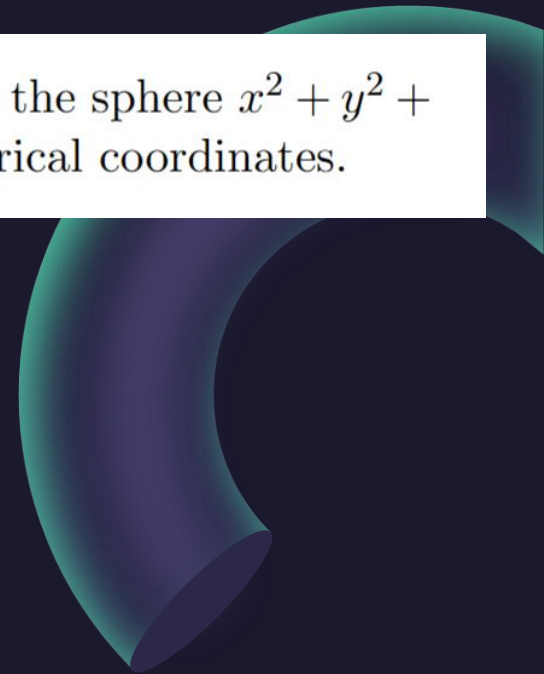
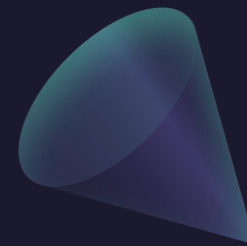


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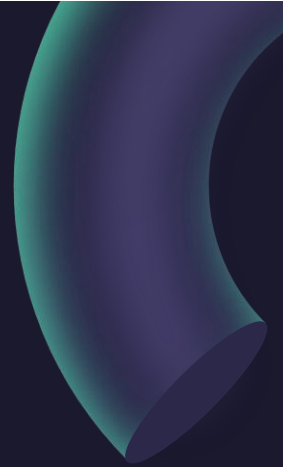
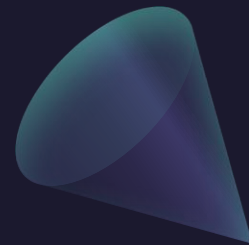
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3. **Spherical Geometry:** Find the volume of the solid E that lies within the sphere $x^2 + y^2 + z^2 = 25$ and above the xy -plane ($z \geq 0$). Set up the integral using spherical coordinates.



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4. **Jacobian Determinant:** Calculate $|J|$, the absolute value of the jacobian, for the transformation: $x = 4u + v, y = u - 2v$.

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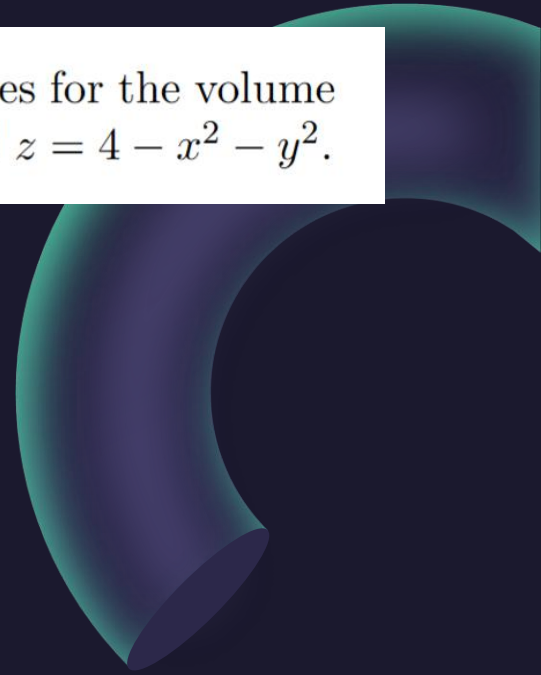
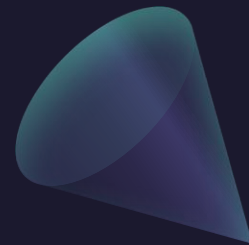
5. **Evaluating Double Integrals:** Evaluate $\iint_D (3x - y) dA$, where D is the triangular region with vertices $(0, 0)$, $(2, 0)$, and $(2, 4)$.

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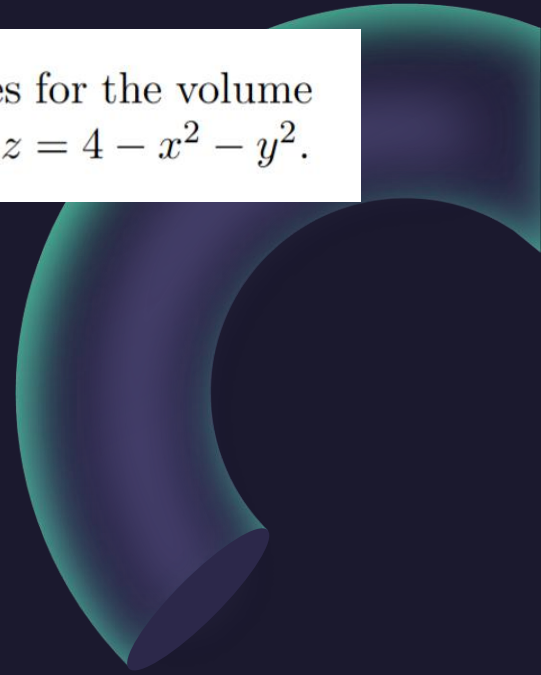
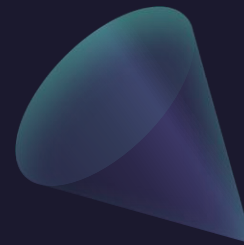
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6. **Cylindrical Coordinates:** Set up a triple integral in cylindrical coordinates for the volume of the solid E bounded below by the xy -plane and above by the paraboloid $z = 4 - x^2 - y^2$.



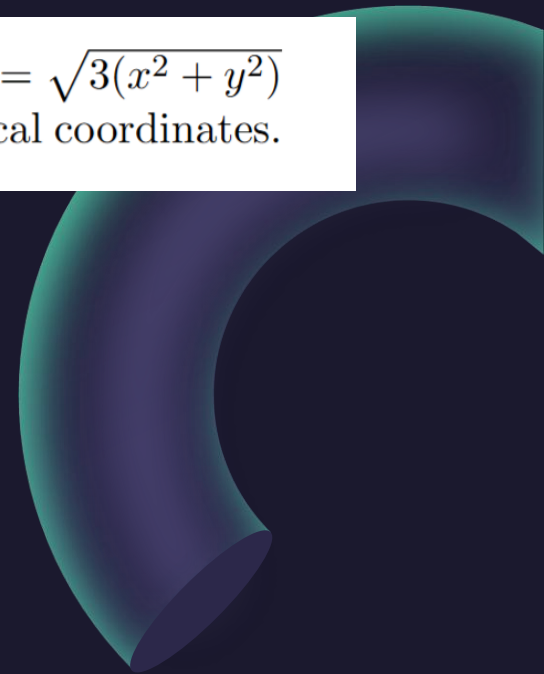
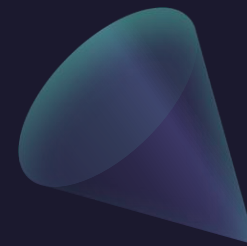
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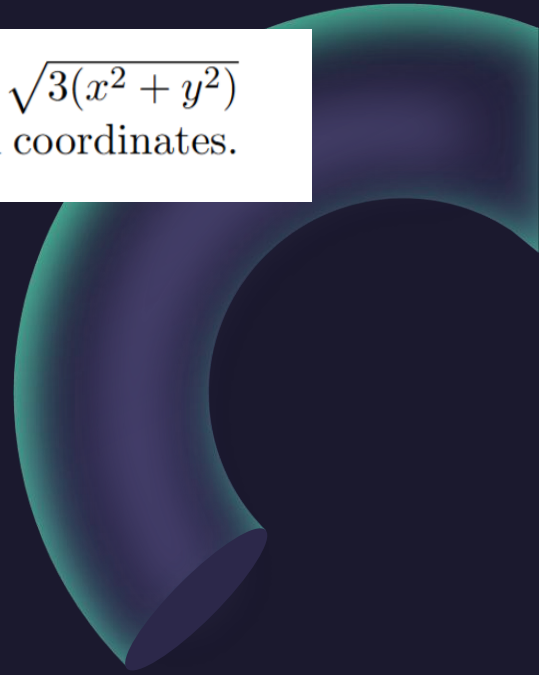
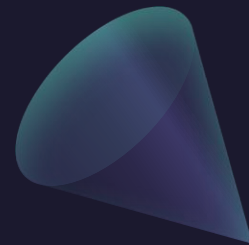
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7. **The Ice Cream Cone:** Let E be the solid region that lies above the cone $z = \sqrt{3(x^2 + y^2)}$ and inside the sphere $x^2 + y^2 + z^2 = 4$. Set up the integral $\iiint_E z^2 dV$ in spherical coordinates.



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Example:

8. **Change of Variables:** Let R be the parallelogram in the xy -plane bounded by the lines

$$x + y = 1, \quad x + y = 3, \quad x - y = 0, \quad x - y = 2.$$

Evaluate the integral $\iint_R (x + y) dA$ by using the linear transformation

$$T(u, v) = (x, y), \quad x = \frac{u + v}{2}, \quad y = \frac{u - v}{2},$$

which transforms R into a square S in the uv -plane.

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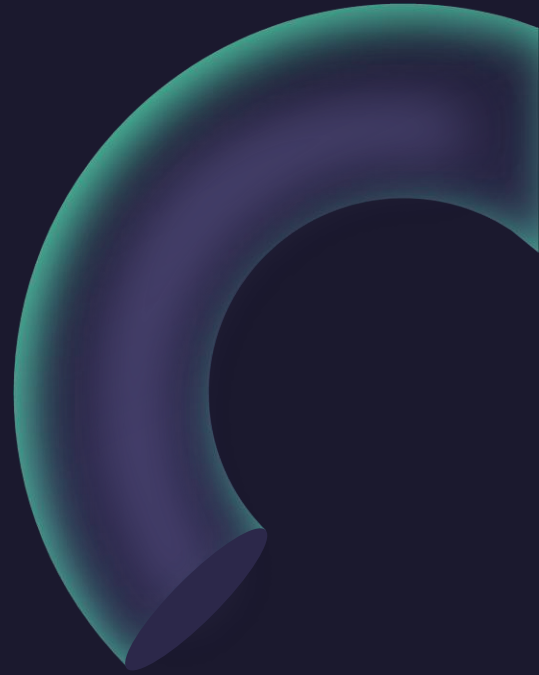
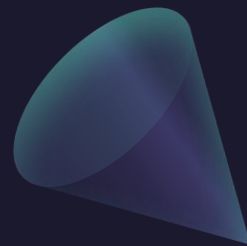
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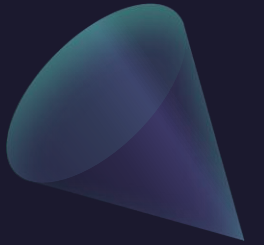
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Questions?



Thank you

Until next time.

